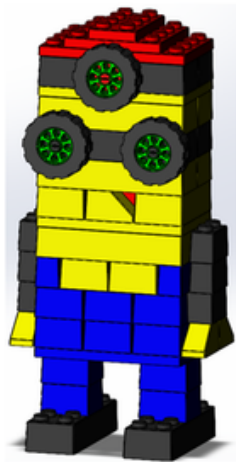
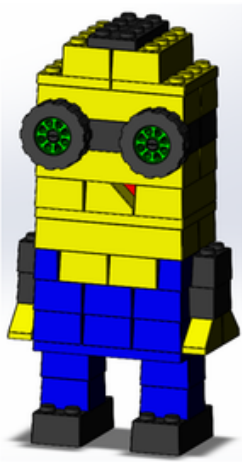


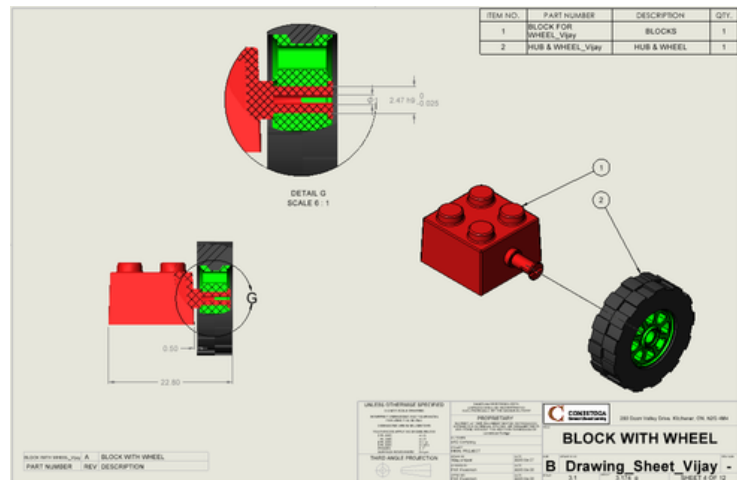
MINIONS - LEGO BLOCKS BUILT



3 – Eyed LEGO Minion



2 – Eyed LEGO Minion



What?

- Designed the individual LEGO block to build a minion using assembly and its features.
- Fun SolidWorks project with all the necessary minor details put in modeling.

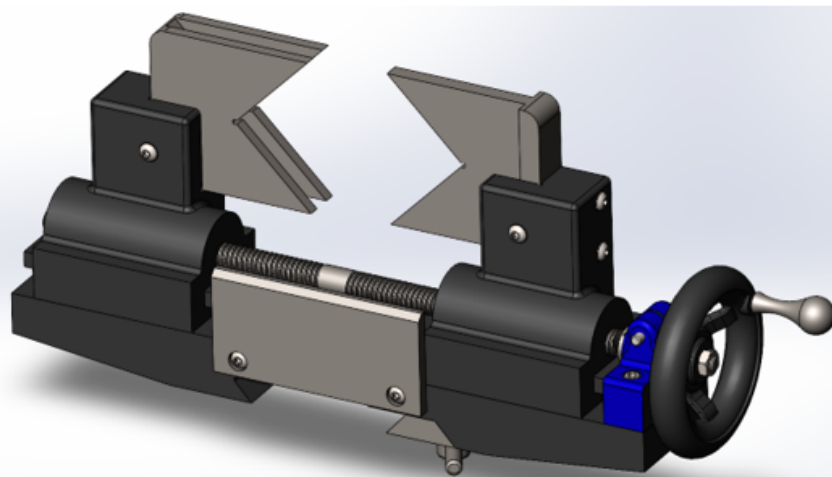
How?

- Used 3D solid modeling features in Solidworks to model this.
- Used essential features such as Configurations, Built-in Design table, and more.

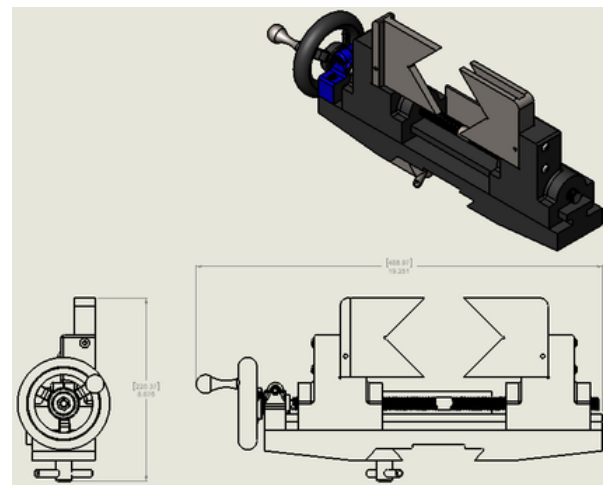
Results

- This minion design has 2 configurations built to use advanced SolidWorks features, such as assembly features, exploded views, and more
- Please click [here](#) to view detailed drawing sheets of this project.

CENTERING ATTACHMENT / VISE - WORKING MODEL



Centering Attachment Device.



What?

- Modeled the centering Attachment / Vise.
- To demonstrate the use of advanced assembly feature and mates.

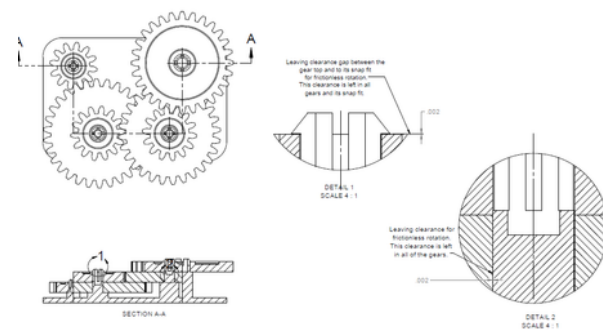
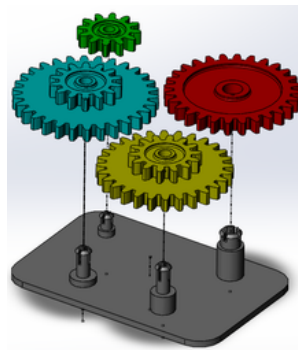
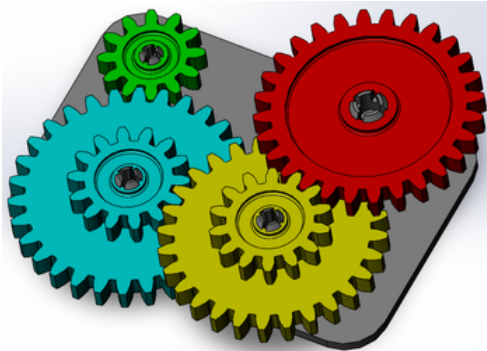
How?

- Designed on SolidWorks.
- Individual components are modeled and put together using assemblies and sub-assembly.
- Created ACME thread specification for the center rod.
- Used Built-in Toolbox Components.

Results

- Working based Centering attachment model using CAD Software.
- Demonstrated the use of advanced assembly features and use of GD&T.

COMPOUND GEAR TRAIN - ASSEMBLY



What?

- Designed a Compound Gear Train.
- Mainly designed for 3D Printing the parts individually.

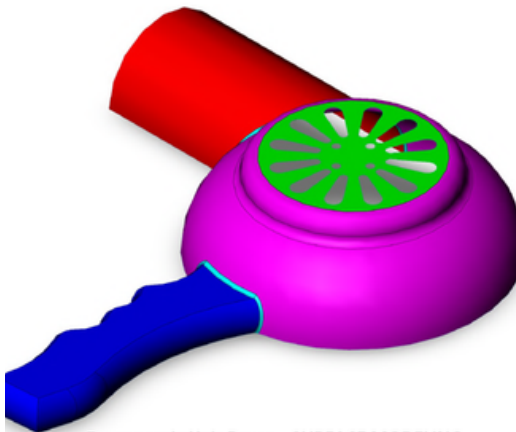
How?

- Used SolidWorks to design the gears
- Implemented the DFA principles and tolerances between parts for movements.
- Used involute equations in gear tooth design.

Results

- Designing for 3D printing demonstrated.
- Implementing DFMA principles for manufacturing and assembly.

MY OTHER SOLIDWORKS PROJECTS:



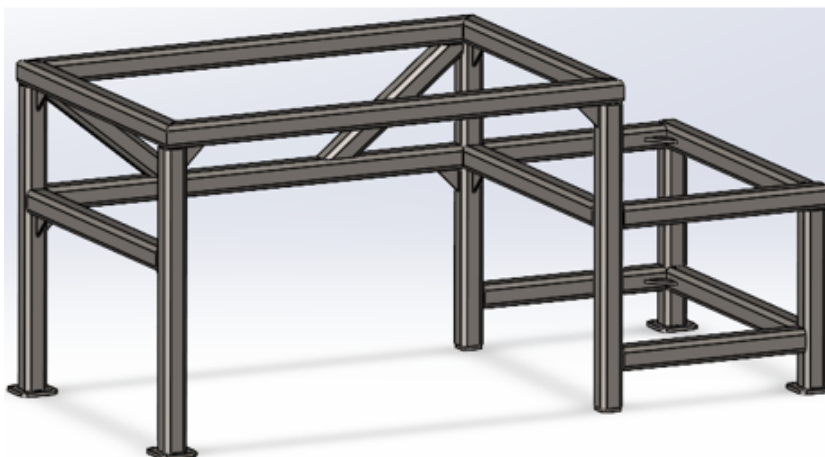
Ergonomic Hair Dryer – SURFACE MODELING.



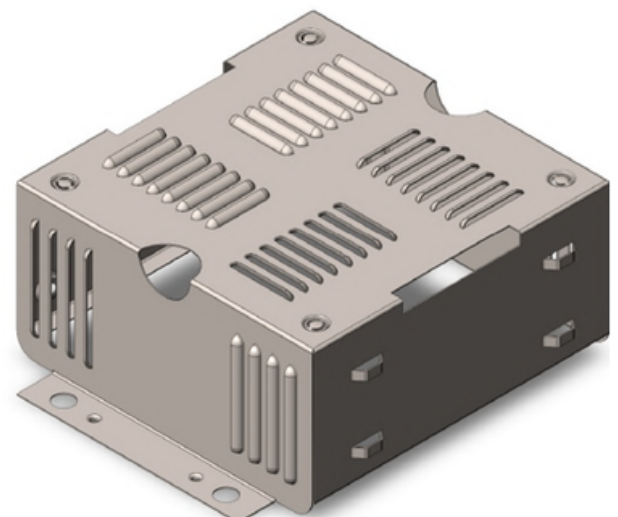
Phone Cover Design



Free Form Surface Modeling.

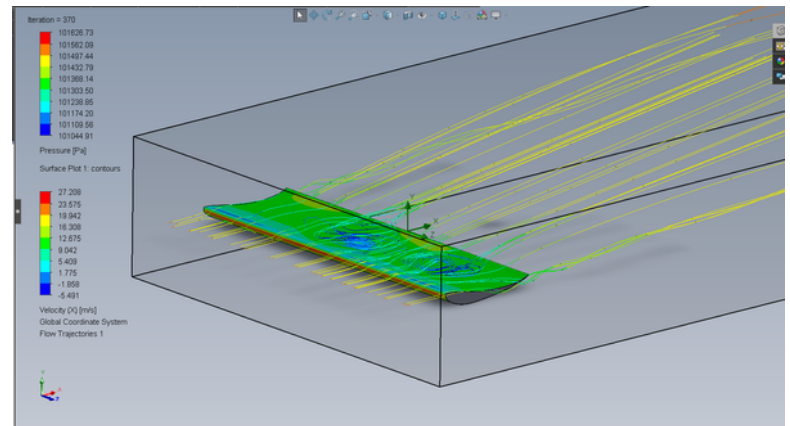
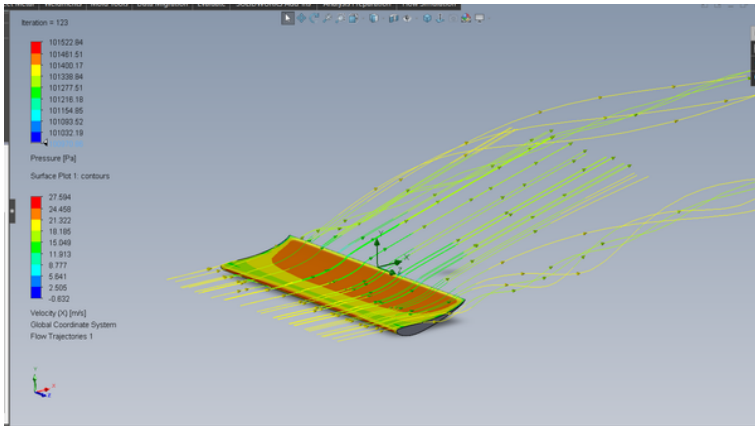


Weldment Design



Sheetmetal Design

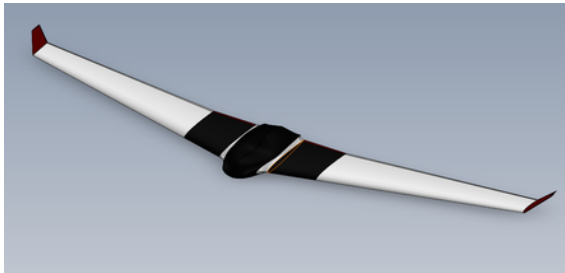
SOLIDWORKS FLOW SIMULATIONS:



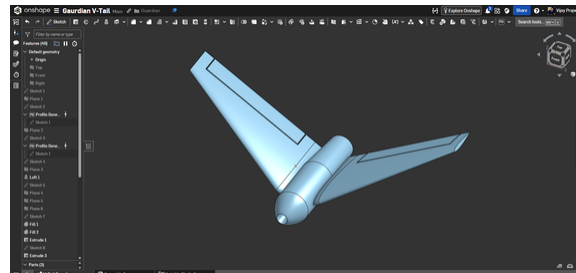
What?

- S1233 Airfoil - FSAE Rear Wing Flow Simulation.
- Ran a Flow Simulation study in SolidWorks on an S1233 airfoil to understand the airflow around the rear-wing section designed for an FSAE car.
- Used pressure contours and flow trajectories to look at how the pressure changes over the airfoil as it moves around it, with the main goal of understanding how the wing can generate downforce for improved traction and cornering on track.

ONSHAPE & SOLIDWORKS:



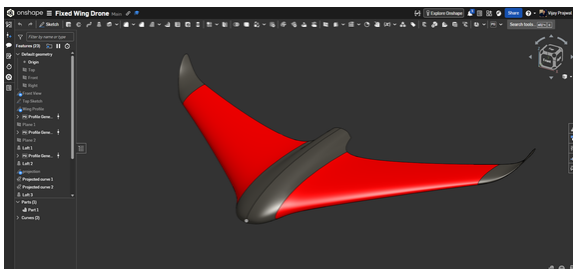
SolidWorks Software: UAV



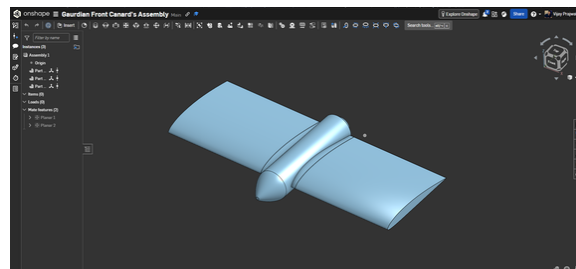
Some designs I worked on at UBCO Aerospace club V-Tail config



SolidWorks Rendering



Onshape Software: UAV



Part of UBCO Aerospace Club - Front Canard's design

